Centre Number	Candidate Number	Name

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

PHYSICS **5054/03**

Paper 3 Practical Test

May/June 2004

2 hours

ANSWER BOOKLET

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen in the spaces provided on this Answer Booklet.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
All of your answers should be written in this Answer Booklet: scrap paper must **not** be used.

Answer **all** questions.
Graph paper is provided in this Answer Booklet. Additional sheets of graph paper should be used only if it is necessary to do so.
At the end of the examination, fasten any additional answer paper used securely to this Answer Booklet.

If you have been given a label, look at the details. If any details are incorrect or missing, please fill in your correct details in the space given at the top of this page.

Stick your personal label here, if provided.

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1	
2	
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4	
Total	

This document consists of **7** printed pages and **1** blank page.

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Section A

1 **(a)** determination of t_1

(b) calculation of T_1

(c) determination of t_2

calculation of T_2

(d) uncertainty in the measurement of t_1

(e) conclusion

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2 **(a)** diagram of the circuit that has been set up by the Supervisor

(b) record of I

record of V

(c) calculation of R_1 using $R_1 = V/I$

(d) record of V

record of I

calculation of R_2 using $R_2 = V/I$

(e) conclusion

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3 **(a)** record of θ_1

(b) **(i)** record of θ_2

(ii) record of V_F

(iii) record of V_I

(iv) record of m_I

(c) calculation of the gain in thermal energy of the ice as it melts using

change in thermal energy on melting = $m_I L$,

where L = 336 J/g

(d) **(i)** calculation of the gain in thermal energy of the cold water formed from the ice using

change in thermal energy = mass × specific heat capacity × temperature change

where specific heat capacity of water = 4.2 J/(g K) and 1 cm^3 of water has a mass of 1 g

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(d) (ii) calculation of the loss in the thermal energy of the water that was initially at room temperature

(e) comment on the answers obtained in parts **(c)** and **(d)**

Section B

4 **(b)** record of x

(c) record of y

calculation of d using $d = x - y$

(d) table of values of D, x, y, d, $(d/D)^2$ and $1/D$

(e) using the grid on page 7, plot a graph of $(d/D)^2$ on the y-axis against $(1/D)/(1/\text{m})$ on the x-axis

(f) calculation of S

(g) determination of f using $f = -S/4$

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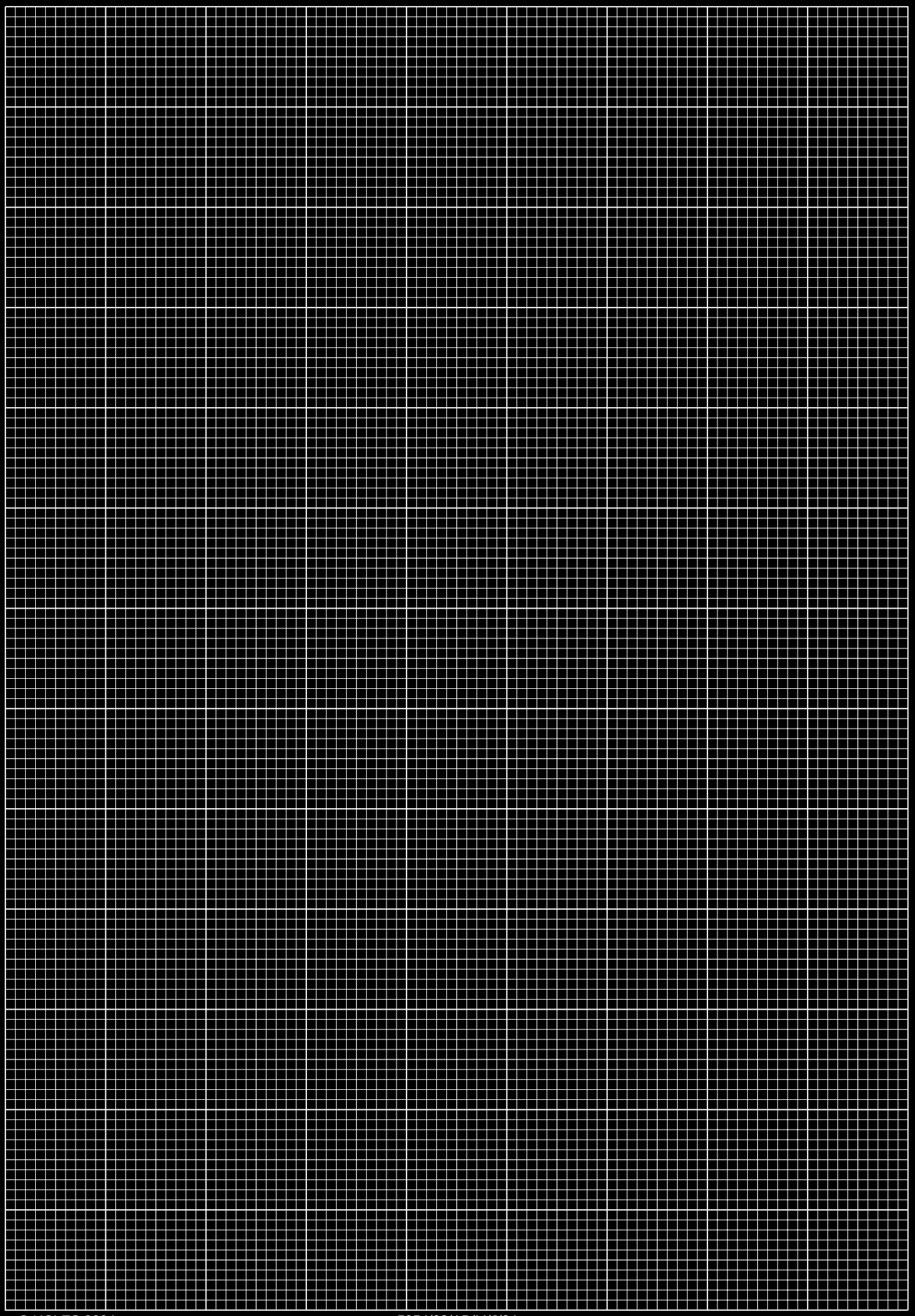

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